JC2001 Introduction to Software Engineering

Practical Session 5 - Exercises

Use Cases

A use case is a methodology used in system analysis to identify, clarify and organise system requirements. A use case consists of a set of possible sequences of interactions between systems and users in a particular environment related to a particular goal. The use case methodology creates a document that describes all the steps a user takes to complete an activity. Use cases add value because they explain how the system should behave and, in the process, help brainstorm what could go wrong. They provide a list of goals that can be used to estimate the system's cost and complexity. Project teams can then negotiate which functions become requirements and are built. Use cases are typically written by business analysts and can be used during several stages of software development, such as planning system requirements, validating design, testing software and creating an outline for online help and user manuals.

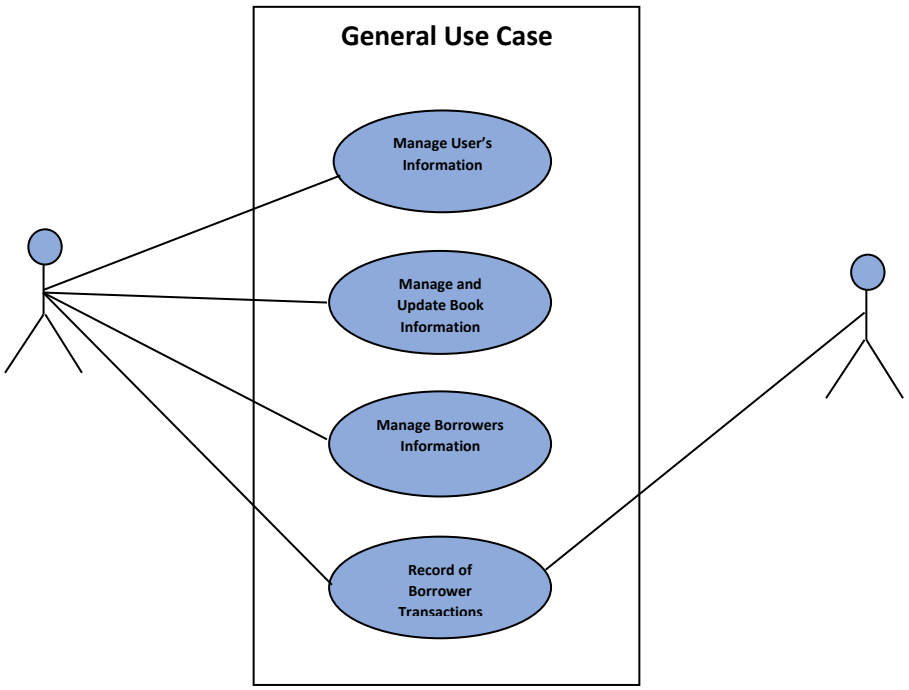
Use cases include (a) who is using the system, (b) what the user wants to do, (c) the user's goal, (d) the steps the user takes to accomplish a particular task and (e) how the system should respond to a user action. Use cases **do not** include (a) implementation-specific language and (b) details about the user interfaces, system hardware or application software. Every use case contains three essential elements:

- **The Actor:** This can be a single person or a group of people interacting with the process.
- **The goal:** The final successful outcome that completes the process.
- **The system:** The process and steps taken to reach the end goal, including the necessary functional requirements and their anticipated behaviours.

Use cases describe a system's functional requirements from the end user's perspective, creating a goal-focused sequence of events that is easy for users and developers to follow. A complete use case will include one main or basic flow and various alternate flows. The alternate flow, also known as an extending use case, describes normal variations to the basic flow as well as unusual situations. A use case should display the following characteristics:

- Organises functional requirements.
- Models the goals of system/actor interactions.
- Records paths – called sequences – from trigger events to goals.
- Describes one main flow of events and various alternate flows.
- Multi-level, so that one use case can use the functionality of another one.

A use case diagram for a library system is illustrated below. It contains two actors, a librarian and a borrower. The librarian manages user information, updates book information, updates borrower information, and records borrower transactions. The borrower is a person who borrows books from the library.



Exercise A: Develop a top-level use case diagram for a Point-of-Sale (POS) system

A Point of Sale refers to a place where a customer pays for goods or services and where sales tax may become payable. It can be in a physical store, where POS terminals and systems process credit card, debit card, or cash payments, or at a virtual sales point such as a computer or mobile electronic device.

Question 1: List and describe all the actors in a point-of-sale.

Question 2: List and describe all the processes involved in executing a sale.

Question 3: Plot a use case diagram for a POS.

Exercise B: Risk Analysis

Activities such as checkout, returns, and refunds are often found in a POS system. The checkout use case is an example of a large, complex use case that can be split into several use cases, each describing a logical unit of behaviour. Note that including a use case becomes incomplete by itself and requires the included use case to be complete.

Question 4: List and describe all the actors in the checkout use case.

Question 5: List and describe all the processes in the checkout use case.

Question 6: Plot a use case diagram for the checkout use case.

References

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3. Use Cases, https://www.usability.gov/how-to-and-tools/methods/use-cases.html